

Baseline-Relative EEG Workload Decoding Replicates in Exploratory Datasets but Fails a Frozen Graded External Transfer Test

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Abstract

EEG workload classifiers often perform well in within-dataset analyses but can fail when tested on unseen people or different task designs. We evaluated a baseline-relative workload decoding protocol using public arithmetic EEG datasets and separated exploratory validation from a frozen external test. In the exploratory datasets, subject-specific z-scoring against resting baseline improved leave-one-subject-out workload decoding relative to absolute features and mean subtraction. With an 8-channel intersection (F3, F4, F7, F8, O1, O2, T3, T4) and 200 per-channel features, z-scored L2 logistic regression achieved MAT window ROC-AUC = 0.865995, and z-scored linear SVM achieved STEW window ROC-AUC = 0.791896. Negative-control permutations were significant for the best MAT and STEW exploratory configurations (empirical $p = 0.000999$ for both). However, feature-direction stability between MAT and STEW was not demonstrated: for z-scored L2 logistic regression, the MAT-STEWE coefficient correlation was $r = 0.0136403$ ($p = 0.847975$). We then froze the best single exploratory pipeline across MAT and STEW (subject z-scoring, median imputation, standard scaling, L2 logistic regression, $C = 1.0$) and applied it without retraining to OpenNeuro DS007262. The confirmatory graded workload test failed: subject-level Spearman correlation between frozen decision score and ordinal difficulty was $\rho = -0.0176745$ with one-sided $p = 0.577865$ across 18 subjects and 1,134 scored trials. These results support baseline-relative calibration as a useful exploratory preprocessing step for within-paradigm EEG workload decoding, but they do not support a universal or monotonic cross-dataset workload axis.

1 Introduction

Cognitive workload is commonly associated with changes in theta- and alpha-band EEG activity during arithmetic, working memory, and attention tasks [1, 2, 4]. These signatures motivate EEG-based workload monitoring, but they do not by themselves establish generalizable prediction. Scalp

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EEG features also encode stable differences between participants, reference montages, electrode layouts, task timing, and recording systems. A model can therefore perform well by exploiting subject- or dataset-specific structure rather than a task-general workload signal [5].

The present manuscript tests the *Baseline-Relative Workload Transfer Hypothesis*: if workload-relevant EEG changes are partly shared across subjects and datasets, then expressing each subject’s task windows relative to that subject’s resting baseline should improve within-dataset decoding and should produce a monotonic increase in frozen model score across graded external workload levels. This hypothesis has two separable parts. The exploratory part asks whether baseline-relative calibration improves classification in datasets with rest-versus-task labels. The confirmatory part asks whether a frozen model trained on exploratory data scales monotonically with difficulty in an untouched graded external dataset.

We evaluated this hypothesis using three public EEG datasets: PhysioNet EEG During Mental Arithmetic Tasks (MAT) [4, 6, 25], STEW [7], and OpenNeuro DS007262 [8]. The analysis was deliberately constrained to results that were actually executed in the repository. We do not report unverified coupling analyses, source reconstruction, or anatomical generator claims, because those analyses were not part of the verified validation pipeline. The central question is methodological: which workload claims survive strict subject-wise validation and a frozen external graded test?

2 Materials and Methods

2.1 Datasets

MAT. The MAT dataset includes 36 participants with resting and mental arithmetic EEG blocks recorded at 256 Hz using a 19-channel 10–20 montage [4, 6]. Files ending in `_1.edf` were treated as rest and files ending in `_2.edf` as arithmetic.

STEW. The STEW dataset includes 48 participants recorded at 128 Hz with a 14-channel Emotiv EPOC+ headset during baseline and mental arithmetic workload conditions [7]. STEW channel names were mapped to the closest standard 10–20 labels where needed.

DS007262. OpenNeuro DS007262 contains arithmetic trials with graded difficulty ranges and 19-channel EEG recorded at 250 Hz [8]. The local event files used in the frozen confirmatory analysis exposed seven non-tutorial numeric difficulty ranges from 0.6–1.5 through 6.0–6.9. The confirmatory analysis scored all available non-tutorial numeric task trials rather than inventing an eighth level.

2.2 Frozen Feature Space

All validation analyses used the same 8-channel intersection across MAT, STEW, and DS007262: F3, F4, F7, F8, O1, O2, T3, and T4. For each channel, 25 per-channel features were used, yielding 200 features total. The feature templates were time-domain moments and morphology, Hjorth parameters, Shannon entropy, spectral entropy, absolute and relative bandpower in delta, theta,

alpha, beta, and gamma bands, and theta/alpha, beta/alpha, and theta/beta ratios. The exact feature definition is locked in `FROZEN_CONFIG.yaml`.

2.3 Exploratory Baseline-Calibration Comparison

For MAT and STEW, the repository implemented strict leave-one-subject-out (LOSO) validation. In each fold, the held-out subject was excluded from calibration statistics, imputation, scaling, and model fitting. The exploratory comparison tested three feature representations:

1. absolute features;
2. mean subtraction relative to subject baseline;
3. z-scoring relative to subject baseline.

MAT used the first 60 seconds of resting windows as calibration rows and excluded those rows from scoring. STEW did not contain timing metadata in the cached parquet table, so the first 50% of each subject’s rest rows by row order were used as calibration rows. L2 logistic regression and linear SVM were evaluated with fixed $C = 1.0$; no label-dependent hyperparameter search was performed in these confirmatory validation scripts.

2.4 Calibration-Duration Curve

For MAT, the resting baseline duration was varied from 1 second upward in 2-second increments. For each duration, the same fixed- C LOSO logistic regression protocol was rerun. The plateau was defined as the first duration within 0.01 ROC-AUC of the maximum valid ROC-AUC. STEW did not support this analysis because the cached table did not include timing metadata.

2.5 Effect-Direction Stability

To test whether a shared signed feature direction was present across exploratory datasets, the fitted coefficient vectors from MAT and STEW models were correlated using Pearson correlation for each model and calibration condition. This analysis tests coefficient-direction stability; it is not a source-localization analysis and cannot establish anatomical invariance.

2.6 Negative Controls

The repository implemented 1,000 within-subject label permutations on nested out-of-fold scores for the best observed MAT and STEW exploratory configurations. The empirical p-value was computed as

$$p = \frac{1 + \#(\text{null AUC} \geq \text{observed AUC})}{1 + N_{\text{valid permutations}}}.$$

These controls test score-label association in the out-of-fold predictions; they do not replace re-trained permutation tests.

2.7 Frozen DS007262 Confirmatory Test

The single model frozen for DS007262 was selected before viewing DS007262 graded scores: subject z-scoring, median imputation, standard scaling, and L2 logistic regression with $C = 1.0$, `solver = liblinear`, `class_weight = balanced`, `max_iter = 5000`, and `random_state = 20260602`. This choice had the best mean exploratory MAT/STEW window ROC-AUC among the tested single-model options. The model was trained on MAT and STEW scored rows only. No DS007262 rows were used for model fitting, model selection, feature selection, or hyperparameter tuning.

For each DS007262 subject, the first 60 seconds after the `started_tutorial_arithmetic` event marker were used as baseline-calibration data and split into non-overlapping 6-second windows. All non-tutorial numeric arithmetic trials with positive duration were scored. The primary confirmatory statistic was Spearman correlation between ordinal workload level and frozen logistic-regression decision score after averaging scores within each subject-level difficulty bin. The prespecified success criterion required at least 12 readable subjects, at least 6 available difficulty levels, positive one-sided Spearman $p < 0.05$, and $\rho \geq 0.20$.

3 Results

3.1 Baseline-Relative Calibration Improved Exploratory Decoding

Z-scoring relative to subject baseline was the strongest calibration strategy in both exploratory datasets (Table 1). In MAT, z-scored L2 logistic regression achieved window ROC-AUC = 0.865995 across 3,186 predictions from 36 subjects. In STEW, z-scored linear SVM achieved window ROC-AUC = 0.791896 across 1,519 predictions from 48 subjects, while z-scored L2 logistic regression was nearly identical at ROC-AUC = 0.791446. Absolute features and mean subtraction were weaker in STEW, and absolute features were weaker than z-scoring in MAT.

Table 1: Exploratory LOSO baseline-calibration comparison using 200 frozen 8-channel features.

Dataset	Model	Calibration	N subjects	N predictions	Window ROC-AUC	Subject AUC mean
MAT	L2 logistic	z-score	36	3186	0.865995	0.874355
MAT	linear SVM	z-score	36	3186	0.858511	0.865624
MAT	L2 logistic	mean subtraction	36	3186	0.826001	0.843782
MAT	L2 logistic	absolute	36	3186	0.736025	0.792007
STEW	linear SVM	z-score	48	1519	0.791896	0.810585
STEW	L2 logistic	z-score	48	1519	0.791446	0.806968
STEW	L2 logistic	absolute	48	1519	0.723308	0.737614
STEW	L2 logistic	mean subtraction	48	1519	0.694381	0.742161

The negative controls were also favorable for the exploratory classifiers. For MAT z-scored L2 logistic regression, observed ROC-AUC was 0.865995, the permutation mean was 0.508327, and empirical $p = 0.000999$. For STEW z-scored linear SVM, observed ROC-AUC was 0.791896, the permutation mean was 0.500369, and empirical $p = 0.000999$.

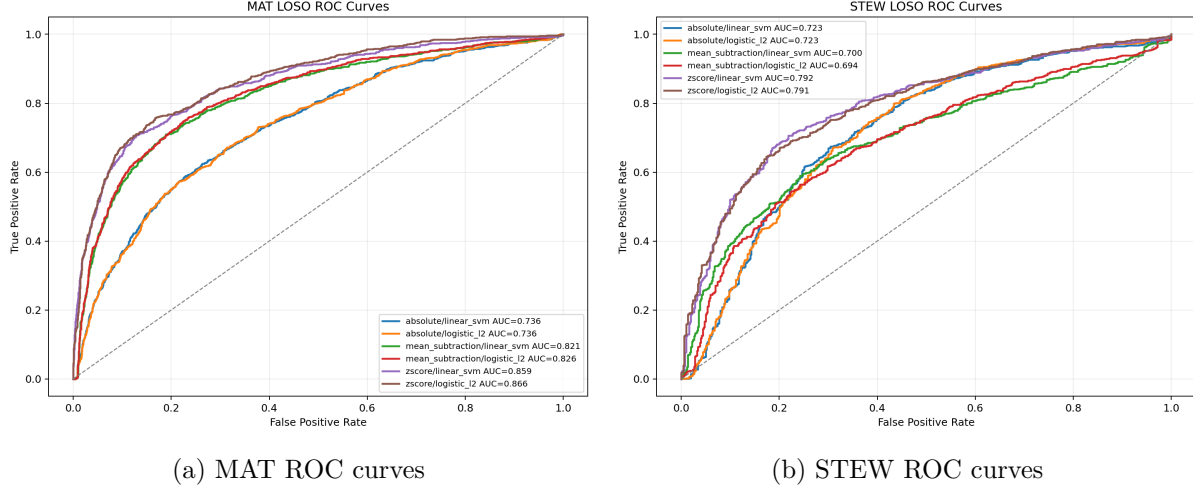


Figure 1: Exploratory LOSO ROC curves from the executed theory-validation pipeline.

3.2 Calibration-Duration Curve

The MAT calibration-duration curve showed a maximum valid ROC-AUC of 0.876133 for z-scored L2 logistic regression. Under the preregistered plateau definition, the first duration within 0.01 ROC-AUC of the maximum was 51 seconds, with plateau ROC-AUC = 0.876133. Durations from 77 seconds onward were not run because too few post-calibration resting windows remained for valid subject-level scoring in all subjects.

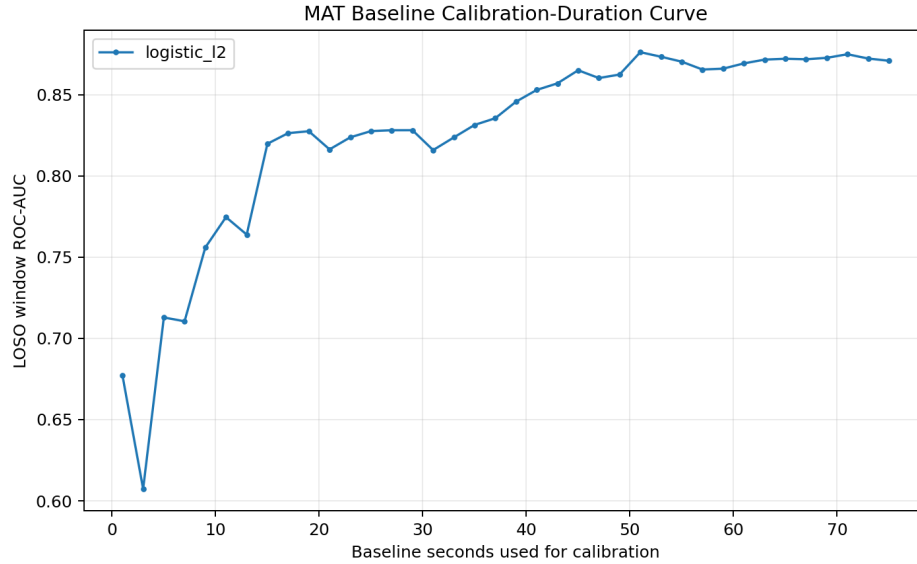


Figure 2: MAT calibration-duration curve from the executed fixed-C LOSO pipeline. The observed plateau occurred at 51 seconds by the rule “first duration within 0.01 ROC-AUC of the maximum valid ROC-AUC.”

3.3 Feature-Direction Stability Was Not Demonstrated

The coefficient-direction analysis did not support a stable shared signed feature axis between MAT and STEW. For the frozen z-scored L2 logistic model, the MAT–STEW coefficient correlation was $r = 0.0136403$ with $p = 0.847975$. Across all tested calibration/model combinations, Pearson correlations ranged from -0.0415691 to 0.0734667 and none were significant (Table 2). This result directly rules out a manuscript claim that the current analyses identified an invariant universal axis.

Table 2: MAT–STEW coefficient-direction stability across exploratory configurations.

Calibration	Model	Pearson r	p-value
absolute	L2 logistic	-0.005644	0.936778
absolute	linear SVM	-0.041569	0.558922
mean subtraction	L2 logistic	0.073467	0.301201
mean subtraction	linear SVM	-0.016498	0.816643
z-score	L2 logistic	0.013640	0.847975
z-score	linear SVM	0.046750	0.510952

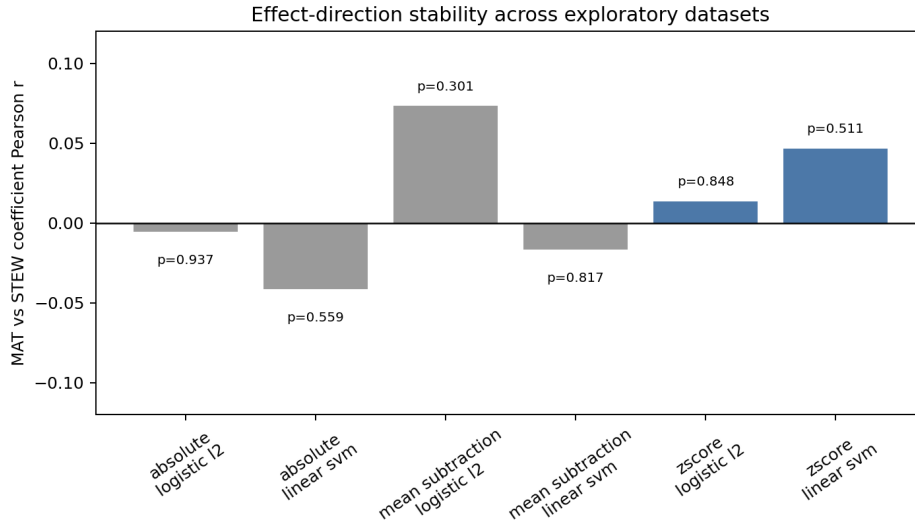


Figure 3: Coefficient-direction stability between MAT and STEW. All correlations were near zero and non-significant, including the frozen z-scored L2 logistic model.

3.4 Frozen DS007262 Graded Transfer Failed

The DS007262 confirmatory test applied the frozen MAT/STEW-trained model without retraining on DS007262. All 18 released subjects were readable, and 1,134 non-tutorial graded task trials were scored. The available event files contained seven ordinal levels, not eight, with 162 trials per level.

The primary confirmatory result was a failure under the locked rules (Table 3). The subject-level Spearman correlation between frozen decision score and ordinal workload level was $\rho = -0.0176745$

with two-sided $p = 0.844270$ and one-sided positive monotonicity $p = 0.577865$. The trial-level Spearman correlation was also negative and non-significant ($\rho = -0.0146273$, one-sided $p = 0.688660$). Per-subject correlations had mean $\rho = -0.0595238$, median $\rho = 0$, and 8 of 18 subjects showed positive correlations.

Table 3: Frozen DS007262 graded workload confirmatory result.

Status	N subjects	N trials	Levels	Subject-level ρ	One-sided p	Trial-level ρ
Failure	18	1134	7	-0.0176745	0.577865	-0.0146273

Table 4: DS007262 frozen score summary by available ordinal workload level.

Difficulty range	Level	Mean score	Median score	N trials
0.6–1.5	1	-1.91867	-2.05471	162
1.5–2.4	2	-2.55800	-2.17073	162
2.4–3.3	3	-3.23046	-2.59551	162
3.3–4.2	4	-2.65596	-3.08935	162
4.2–5.1	5	-3.40334	-2.73853	162
5.1–6.0	6	-2.82036	-2.62951	162
6.0–6.9	7	-2.44009	-2.35241	162

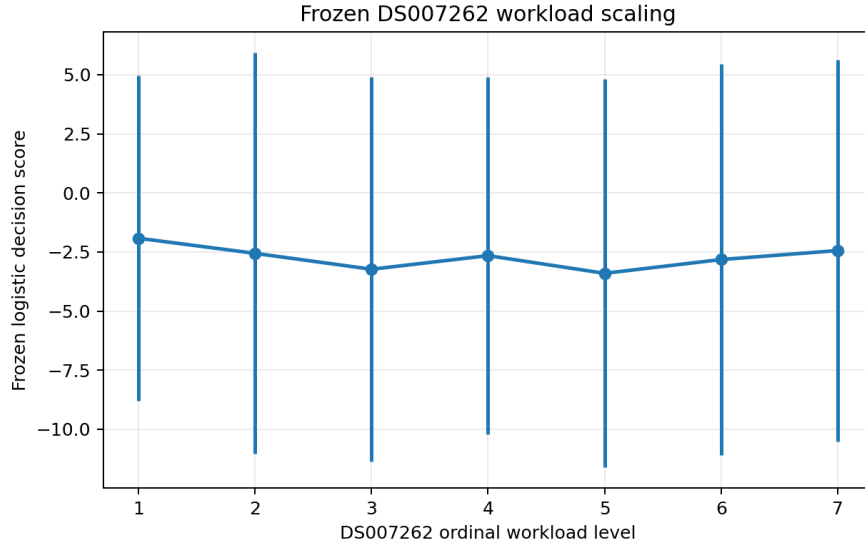


Figure 4: Frozen DS007262 graded workload scores. The fitted exploratory MAT/STEW model did not produce a positive monotonic score increase across the available DS007262 difficulty levels.

4 Discussion

4.1 What the Verified Analyses Support

The verified analyses support a limited but useful conclusion: subject-level baseline z-scoring improved exploratory workload decoding in MAT and STEW relative to absolute features and mean subtraction. The effect was strongest in MAT and was also present in STEW. Negative controls showed that the observed out-of-fold scores were associated with labels in the exploratory datasets.

The analyses do not support a stronger claim that the model recovered a universal or invariant workload axis. The coefficient-direction correlations between MAT and STEW were near zero. Most importantly, the frozen DS007262 graded test failed the prespecified monotonicity criterion. A model can therefore classify rest-versus-workload labels in exploratory datasets while still failing to rank graded workload levels in an external dataset.

4.2 Interpretation of the Baseline-Relative Workload Transfer Hypothesis

The Baseline-Relative Workload Transfer Hypothesis was partly supported and partly falsified. The exploratory part was supported: baseline-relative z-scoring improved LOSO decoding in MAT and STEW. The confirmatory transfer part was not supported: the frozen model did not produce a positive monotonic relationship with DS007262 difficulty. The correct interpretation is not that baseline-relative calibration is useless, but that the current implementation does not establish graded cross-dataset transfer.

Several non-exclusive explanations are compatible with the failure, but the present analyses do not adjudicate among them. DS007262 differs from MAT and STEW in task structure, event timing, difficulty design, and recording context. The channel intersection may omit useful electrodes. Rest-versus-task classification may also differ from ranking multiple arithmetic difficulty levels. These are methodological boundary conditions, not verified neurobiological mechanisms.

4.3 Resting Phenotyping

Exploratory resting EEG predictors should not be treated as a validated baseline phenotype. The prior audit found that no MAT resting feature survived FDR correction at $q = 0.05$ across the tested predictors, and the resting-screen model did not establish a reliable subject-level prediction rule. Resting EEG may still be useful for calibration, as shown by the z-scoring results, but the current repository does not support screening participants or explaining low classifiability by a specific resting-state biomarker.

4.4 Limitations

First, MAT and STEW were used for exploratory model selection and should not be interpreted as independent confirmation. Second, DS007262 provided only seven non-tutorial difficulty ranges in the analyzed event files, although the dataset is described as an 8-level arithmetic task. The

analysis reported the available levels rather than relabeling or imputing an absent level. Third, the negative-control permutations were applied to nested out-of-fold scores and did not retrain models for every null draw. Fourth, coefficient stability was tested only at the sensor-feature level; no source localization or anatomical inference was performed. Fifth, unverified oscillatory coupling analyses were removed because they were not part of the executed confirmatory evidence.

5 Conclusion

Baseline-relative EEG calibration improved exploratory rest-versus-workload decoding in MAT and STEW, with z-scored models achieving ROC-AUC values of 0.865995 and 0.791896 in the best exploratory configurations. However, the feature-direction analysis did not demonstrate a stable shared signed axis, and the frozen DS007262 graded external test failed ($\rho = -0.0176745$, one-sided $p = 0.577865$). The manuscript therefore supports baseline-relative calibration as an exploratory method, but it does not support claims of invariant workload transfer or universal EEG workload geometry.

6 Data and Code Availability

The original datasets are available from PhysioNet EEGMAT [6], STEW [7], and OpenNeuro DS007262 [8]. The analysis code, frozen configuration, and generated validation outputs are stored in the project repository. The key executed summaries are `THEORY_VALIDATION_RESULTS.md` and `results/ds007262_confirmatory/DS007262_CONFIRMATORY_RESULTS.md`.

7 Ethics Statement

This work is a secondary analysis of publicly available, de-identified EEG datasets. No new human-subject data were collected. The original datasets were collected under their respective ethics approvals and consent procedures. The present work is intended for computational neuroscience and methodological validation, not clinical diagnosis or individual assessment.

8 Author Contributions

Abhijay Gangarapu and Arush Ravipati contributed equally to the conception, analysis, interpretation, and manuscript preparation for this study.

9 Conflict of Interest

The authors declare no commercial or financial relationships that could be construed as a potential conflict of interest.

10 Acknowledgments

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